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Roll no: 3253

Division: C

Subject: SS

Assignment – 2

Q1.

```
#include<stdio.h>
```

```
#include<conio.h> #include<ctype.h>
```

```
#include<string.h>
```

```
char stat_table[6][4][10]={  
    {"stat","latter","digit","."},  
    {"start","id","int","error"},  
    {"id","id","id","error"},  
    {"int","error","int","s"},  
    {"s","error","real","error"},  
    {"real","error","real","error"},  
};
```

```
int main()
```

```
{
```

```

    char input[20],current_stat[30],column_stat[30],next_stat[30];
char ch,choice;

    int c,r,error,i,len;

    do{
        printf("enter
identifiers: ");
scanf("%s",input);    len =
strlen(input);
strcpy(current_stat,"start");
error=0;

        for(i=0;i<len && !error;i++)
        {
            ch= input[i];
if(isalpha(ch))
        {
            strcpy(column_stat,"letter");
        }
        else if(isdigit(ch))
        {
            strcpy(column_stat,"digit");
        }
        else if(ch == '.')
        {
            strcpy(column_stat,".");
        }
        else{
            strcpy(next_stat,"error");

```

```

        error = 1;
        break;
    }
    for(r=1;r<6;r++)
    {
        if(strcmp(stat_table[r][0],current_stat)==0){
            for(c=1;c<4;c++){
if(strcmp(stat_table[0][c],column_stat)==0){
strcpy(next_stat,stat_table[r][c]);
                break;
            }
        }
        if(strcmp(next_stat,"error")==0){
            error = 1;
break;
        }
        printf("%s -> %s \n ",current_stat,next_stat);
strcpy(current_stat,next_stat);
        break;
    }
}

if(error)
{
    printf("\n invalid token");
}
else

```

```

{
    printf("\n valid token"); if(strcmp(current_stat,"id")==0){
        printf("\n It is an identifiers");
    }
    else if(strcmp(current_stat,"int")==0){
        printf("\n it is an integer");
    }
    else if(strcmp(current_stat,"real")==0){
        printf("\n it is a real");
    }
}

printf("\n\n Do you want to continue?(enter 'y' for yes and 'n'for no): ");
scanf(" %c",&choice); }while(choice !='n'); return 0;
}

```

```

FPS 0
Name: kavish rathod
Roll no: 3253
enter identifiers: 111.21
start -> int
int -> int
int -> int
int -> s
s -> real
real -> real

valid token
it is a real

Do you want to continue?(enter 'y' for yes and 'n'for no): y

enter identifiers: 12.1.1
start -> int
int -> int
int -> s
s -> real

invalid token

Do you want to continue?(enter 'y' for yes and 'n'for no): n_

```

Q2.

```

#include<stdio.h>

#include<conio.h>

#include<string.h>

```

```
#include<ctype.h>
```

```
char kw[32][10]={"int","float","return","for","char","break"};
```

```
char op[15]={'+','-','*','/','=',':','<','>','&','%','^'};
```

```
char identifiers[20][10]; char
```

```
constants[20][10];
```

```
int ic=0,cc=0;
```

```
void analyzeString(char str[]);
```

```
int main()
```

```
{
```

```
FILE *file; char
```

```
str[100];
```

```
clrscr(); file =
```

```
fopen("input.txt","r");
```

```
if(file== NULL)
```

```
{
```

```
    printf("error opening the file");
```

```
    getch(); return 1;
```

```
}
```

```
while(fgets(str,sizeof(str),file)!= NULL){
```

```
    analyzeString(str)
```

```
; } fclose(file);
```

```
getch(); return 0;
```

```
}
```

```

void analyzeString(char str[])
{
    char *ptr;
    int i,j;

    ptr = strtok(str," \n");
    while(ptr != NULL){
        int flag =0;

        for(i=0;i<32;i++)
        {
            if(strcmp(ptr,kw[i])==0)
            {
                printf("KW#%d \t",i+1);
                flag=1;
            }
        }
        break;

        if(flag == 0){
            for(j=0;j<10;j++){
                if(ptr[0]==op[j]){
                    printf("OP#%d \t",j+1);
                    flag=1;
                    break;
                }
            }
        }
    }
}

```

```

        if (flag == 0){            if(isalpha(ptr[0])){
int isRepeated=0;
for(i=0;i<ic;i++){
if(strcmp(ptr,identifiers[i])==0){
printf("ID#%d \t",i+1);
isRepeated=1;
                break;
            }
        }
        if(!isRepeated){
strcpy(identifiers[ic++],ptr);
printf("ID#%d \t",ic);
        }
    }
    else if(isdigit(ptr[0])){            int
isRepeated=0;            for(i=0;i<cc;i++){
if(strcmp(ptr,constants[i])==0){
                printf("CO#%d \t",i+1);
isRepeated = 1;
                break;
            }
        }
        if(!isRepeated){            strcpy(constants[cc++],ptr);
printf("CO#%d \t",cc);
        }
    }
}
ptr = strtok(NULL, " \n");

```

```
}  
}
```

FPS 0

Name: kavish rathod

Roll no: 3253

KW#1	ID#1	KW#4	ID#2	OP#5	CO#1	ID#3	OP#5	CO#2	ID#2
OP#8	CO#3	ID#2	OP#1	ID#3	CO#4	ID#3	OP#3	CO#5	ID#2
ID#3	CO#6	ID#3	OP#2	CO#7	KW#3	CO#8	-		

Q3.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
#include <conio.h>
```

```
// List of keywords char
```

```
kw[11][10] = {
```

```
    "int", "float", "while", "for", "do", "char", "double", "if", "else", "void", "return"
```

```
};
```

```
// Global array for storing identifiers
```

```
char identifiers[20][10]; int ic = 0;
```

```
// Counter for identifiers
```



```
void analyzeString(char str[]);
```

```
int main() {
```

```
FILE *file;
```

```
    char str[200]; // Buffer to read each line
```

```
    clrscr();
```

```
    printf("name : Kavish rathod
```

```
\n");    printf("Roll no: 3253 \n");
```

```
file = fopen("input.txt", "r");    if
```

```
(file == NULL) {    printf("\n Error
```

```
opening file.\n");    getch(); //
```

```
Wait for a key press    return 1;
```

```
    }
```

```
    // Read and process each line of the file
```

```
while (fgets(str, sizeof(str), file) != NULL) {
```

```
analyzeString(str);
```

```
    }
```

```
        fclose(file);    getch(); // Wait for a key press before closing  
the console window return 0;
```

```
}
```

```
void analyzeString(char str[]) {
```

```
    char *ptr;
```

```
    int i;
```

```
    /*
```

Tokenize the line using a set of delimiters that removes whitespace and common punctuation: space, newline, comma, semicolon, parentheses, and braces.

```
*/ ptr = strtok(str, "\n,;(){}"); while (ptr !=
NULL) {
    int flag = 0;
    // Check if the token is a keyword
    for (i = 0; i < 32; i++) {
if (strcmp(ptr, kw[i]) == 0) {
printf("KW#%d ", i + 1);
        flag = 1;
break;
    }
}

    // If not a keyword and it begins with an alphabet, treat it as an identifier
if (!flag && isalpha(ptr[0])) { int found = 0; for (i = 0; i < ic; i++) {
if (strcmp(ptr, identifiers[i]) == 0) { printf("ID#%d ", i + 1);
        found = 1;
break;
    }
}

    if (!found) {
strcpy(identifiers[ic], ptr);
        ic++;
        printf("ID#%d ", ic);
    }
}

    ptr = strtok(NULL, "\n,;(){}");
```

```
}  
}
```

```
FPS 0 name : kavish rathod  
Roll no: 3253  
KW#1 ID#1 KW#4 KW#1 ID#2 ID#3 ID#2 ID#2 ID#3 ID#3 ID#2 ID#3 ID#3 KW#11
```